Q1 JEE Main 2020 - 2 September (Morning)

A $5\mu\mathrm{F}$ capacitor is charged fully by a $220\mathrm{V}$ supply. It is then disconnected from the supply and is connected in series to another uncharged $2.5\mu\mathrm{F}$ capacitor. If the energy change during the charge redistribution is $\frac{\mathrm{X}}{100}\mathrm{J}$ then value of X to the nearest integer is ________.

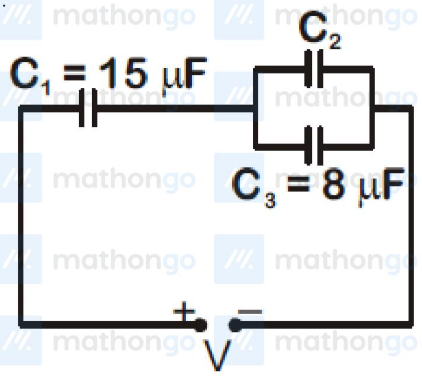
Q2 JEE Main 2020 - 2 September (Evening)

$A10\mu F$ capacitor is fully charged to a potential difference of $50\mathrm{V}$. After removing the source voltage it is connected to an uncharged capacitor in parallel. Now the potential difference across them becomes $20\mathrm{V}$. The capacitance of the second capacitor is

(A) $20\mu\mathrm{F}$
(B) $10\mu\mathrm{F}$
(C) $15\mu\mathrm{F}$
(D) $30\mu\mathrm{F}$

Q3 JEE Main 2020 - 3 September (Morning)

In the circuit shown in the figure, the total charge is $750\mu\mathrm{C}$ and the voltage across capacitor C_2 is $20\mathrm{V}$. Then the charge on capacitor C_2 is



(A) $650\mu\mathrm{C}$
(B) $590\mu\mathrm{C}$
(C) $160\mu\mathrm{C}$
(D) $450\mu\mathrm{C}$

Q4 JEE Main 2020 - 4 September (Evening)

A capacitor C is fully charged with voltage V_0. After disconnecting the voltage source, it is connected in parallel with another uncharged capacitor of capacitance $\frac{\mathrm{C}}{2}$. The energy loss in the process after the charge is distributed between the two capacitors is

(A) $\frac{1}{4}CV_0^2$
(B) $\frac{1}{3}\mathrm{CV}_0^2$
(C) $\frac{1}{6}\mathrm{CV}_0^2$
(D) $\frac{1}{2}CV_0^2$

Q5 JEE Main 2020 - 5 September (Morning)

Two capacitors of capacitances C and $2\mathrm{C}$ are charged to potential differences V and $2\mathrm{V}$, respectively. These are then connected in parallel in such a manner that the positive terminal of one is connected to the negative terminal of the other. The final energy of this configuration is

(A) $\frac{3}{2}\mathrm{CV}^2$
(B) $\frac{9}{2}\mathrm{CV}^2$
(C) Zero
(D) $\frac{25}{6}\mathrm{CV}^2$

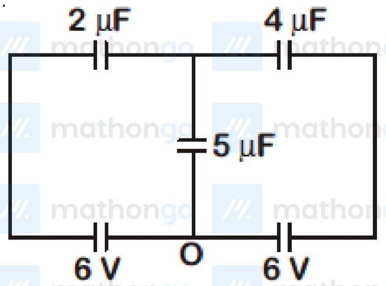
Q6 JEE Main 2020 - 5 September (Evening)

A parallel plate capacitor has plate of length is connected to a battery of emf V. A dielectric slab of the same thickness 'd' and of dielectric constant $\mathrm{k}=4$ is being inserted between the plates of the capacitor. At what length of the slab inside plates, will the energy stored in the capacitor be two times the initial energy stored?

(A) $\frac{\ell}{4}$
(B) $\frac{\ell}{2}$
(C) $\frac{2\ell}{3}$
(D) $\frac{\ell}{3}$

Q7 JEE Main 2020 - 5 September (Evening)

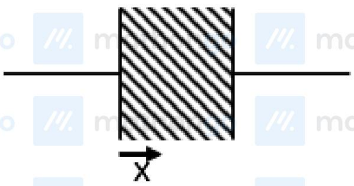
In the circuit shown, charge on the $5\mu\text{F}$ capacitor is



(A) $16.36\mu\text{C}$
(B) $18.00\mu\text{C}$
(C) $5.45\mu\text{C}$
(D) $10.90\mu\text{C}$

Q8 JEE Main 2020 - 7 January (Morning)

A parallel plate capacitor has plates of area A separated by distance d between them. It is filled with a dielectric which has a dielectric constant that varies as $k(x)$=$k_o(1+\alpha x)$ where x is the distance measured from one of the plates. If (αd) < < 1, the total capacitance of the system is best given by the



expression:

(A) $\frac{k_0\varepsilon_0 A}{d}\left(1+\alpha^2 d^2\right)$
(B) $\frac{k_0\varepsilon_0 A}{d}\left(1+\frac{\alpha d}{2}\right)$
(C) $\frac{k_0\varepsilon_0 A}{2d}(1+\alpha d)$
(D) $\frac{k_0\varepsilon_0 A}{2d}\left(1+\frac{\alpha d}{2}\right)$

Q9 JEE Main 2020 - 7 January (Evening)

A $60pF$ capacitor is fully charged by a $20V$ supply. It is then disconnected from the supply and is connected to another uncharged $60pF$ capacitor in parallel. The electrostatic energy that is lost in this process by the time the charge is redistributed between them is (in nJ)

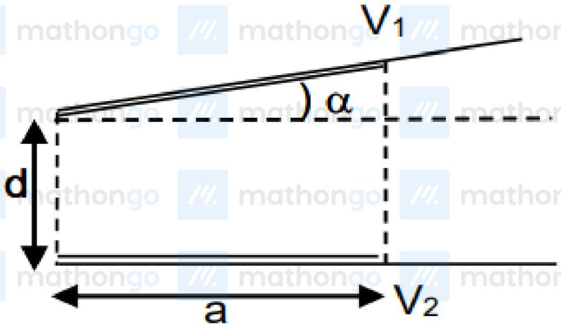
Q10 JEE Main 2020 - 8 January (Morning)

Effective capacitance of parallel combination of two capacitors C_1 and C_2 is $10\mu F$. When these capacitors are individually connected to a voltage source of $1V$, the energy stored in the capacitor C_2 is 4 times that of C_1 . If these capacitors are connected in series, their effective capacitance will be:

(A) $1.6\ \mu F$
(B) $16\ \mu F$
(C) $4\ \mu F$
(D) $\frac{1}{4}\ \mu F$

Q11 JEE Main 2020 - 8 January (Evening)

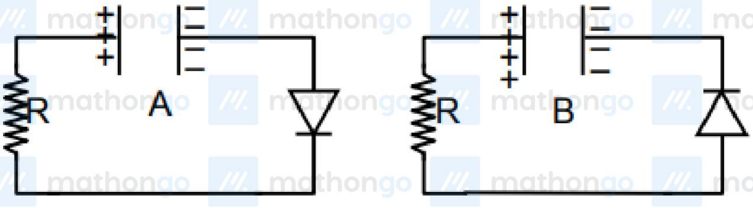
A capacitor is made of two square plates each of side a making a very small angle α between them, as shown in figure. The capacitance will be close to :



(A) $\frac{\varepsilon_0 a^2}{d}\left(1-\frac{\alpha a}{2d}\right)$
(B) $\frac{\varepsilon_0 a^2}{d}\left(1-\frac{\alpha a}{d}\right)$
(C) $\frac{\varepsilon_0 a^2}{d}\left(1+\frac{\alpha a}{2d}\right)$
(D) $\frac{\varepsilon_0 a^2}{d}\left(1-\frac{\alpha a}{4d}\right)$

Q12 JEE Main 2020 - 9 January (Evening)

Two identical capacitors A and B, charged to the same potential $5\ V$ are connected in two different circuits as shown below at time $t=0$. If the charge on capacitors A and B at time $t=CR$ is Q_A and Q_B respectively, then (Here e is the base of natural logarithm)

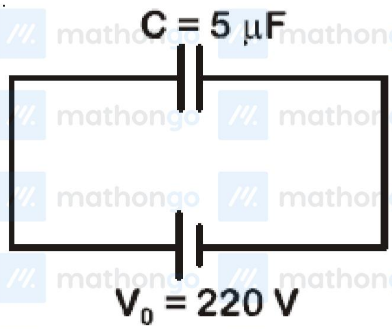


(A) $Q_A = VC,\ Q_B\ CV$
(B) $Q_A = VC,\ Q_B = VC/e$
(C) $Q_A = VC/e,\ Q_B = CV/2$
(D) $Q_A = CV/2,\ Q_B = VC/e$

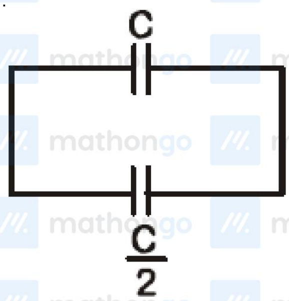
Answer Key

Q1 (4)	**Q2** (C)	**Q3** (B)	**Q4** (C)
Q5 (A)	**Q6** (D)	**Q7** (A)	**Q8** (B)
Q9 (6)	**Q10** (A)	**Q11** (A)	**Q12** (B)

Q1



$Q_0 = CV_0$
Now battery is disconnected and a capacitor of $\frac{\mathrm{C}}{2}$ is connected to C then



Total charge $= \mathrm{Q}_0$
$C_{eq} = \frac{3C}{2}$
$\therefore \quad \Delta E = \frac{Q_0^2}{2C} - \frac{2Q_0^2}{3C \times 2} \Rightarrow \Delta E = \frac{Q_0^2}{2C}\left(\frac{1}{3}\right)$
$\Rightarrow \quad \Delta E = \frac{1}{6} \times 220 \times 220 \times 5 \times 10^{-6} \approx 4 \times 10^{-2} J$
$\therefore \quad X = 4$

Q2

$V_{\text{final}} = \frac{C_1 V}{C_1 + C_2}$
$\mathrm{C}_2 = 15 \mu\mathrm{F}$

Q3

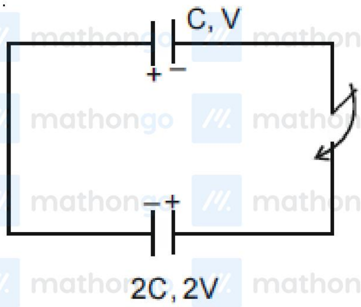
$Q_3 = C_3V_3$
$V_3 = 20\text{V}$
$\text{Q}_3 = 160\mu\text{C}$
$\text{Q} = \text{Q}_2 + \text{Q}_3$
$\text{Q}_2 = 590\mu\text{C}$

Q4

$\Delta\mathbf{U} = \frac{1}{2}\frac{\text{C}_1\text{C}_2}{\text{C}_1+\text{C}_2}\text{V}_0^2 = \frac{1}{6}\text{CV}_0^2$

Q5

$V_{\text{common}} = \frac{4\text{CV}-\text{CV}}{3\text{C}} = \text{V}$

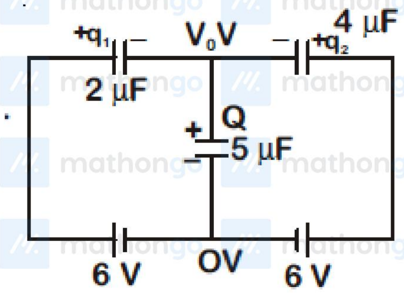


$C_{eq} = C + 2C = 3C$
$\therefore U_f = \frac{1}{2} \times (3C) \times V^2 = \frac{3}{2}CV^2$

Q6

$\text{U} = \frac{1}{2}\text{CV}^2$
$\frac{U_1}{U_2} = 2 \Rightarrow \frac{C_1}{C} = 2$
$\frac{kx+(\ell-x)}{\ell} = 2$
$x = \frac{\ell}{3}$

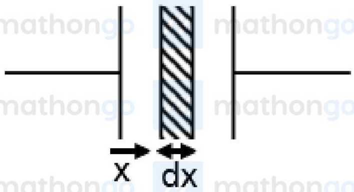
Q7



$$Q = q_1 + q_2$$
$$5V_0 = 2(6 - V_0) + 4(6 - V_0)$$
$$V_0 = \frac{36}{11}V$$
$$Q = 5V_0 = \frac{180}{11}\mu C$$

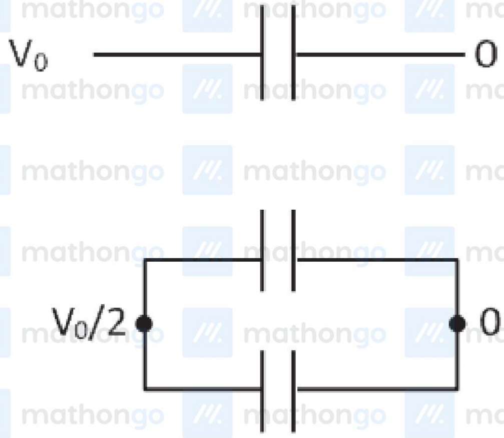
Q8

$$\text{Capacitance of element} = \frac{k\varepsilon_0 A}{dx}$$
$$\text{Capacitance of element } C' = \frac{k_0(1+\alpha x)\varepsilon_0 A}{dx}$$



$$\sum \frac{1}{C'} = \int_0^d \frac{dx}{k_0\varepsilon_0 A(1+\alpha x)}$$
$$\frac{1}{C} = \frac{1}{k_0\varepsilon_0 A\alpha}\ell n(1+\alpha d)$$
$$\text{Given } -\alpha d << 1$$
$$\frac{1}{C} = \frac{1}{k_0\varepsilon_0 A\alpha}\left(\alpha d - \frac{\alpha^2 d^2}{2}\right)$$
$$\frac{1}{C} = \frac{d}{k_0\varepsilon_0 A}\left(1 - \frac{\alpha d}{2}\right)$$
$$C = \frac{k_0\varepsilon_0 A}{d}\left(1 + \frac{\alpha d}{2}\right)$$

Q9



$V_0 = 20\ V$

Heat loss $= U_i - U_f$

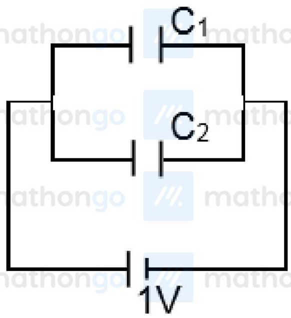
$= \frac{1}{2} C V_0^2 - 2\left[\frac{1}{2} C\left(\frac{V_0}{2}\right)^2\right]$

$= \frac{C V_0^2}{4}$

$= \frac{(60 \times 10^{-12})(20)^2}{4} J$

$= 6 \times 10^{-9} J = 6\ nJ$

Q10



Given $C_1 + C_2 = 10 \mu F \quad \ldots$ (i)

$4\left(\frac{1}{2} C_1 V^2\right) = \frac{1}{2} C_2 V^2$

$\Rightarrow \quad 4 C_1 = C_2 \qquad \ldots$ (ii)

from equation (i) & (ii)

$C_1 = 2 \mu F$

$C_2 = 8 \mu F$

If they are in series

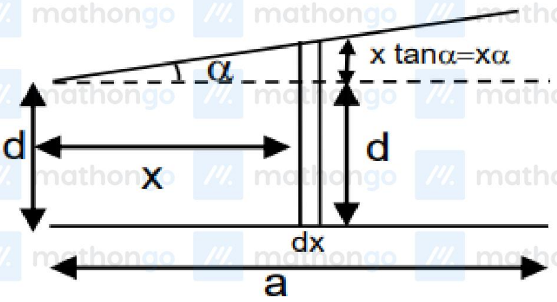
$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = 1.6 \mu F$

Q11

$$dc = \frac{\varepsilon_0 a dx}{d + \alpha x}$$

$$\Rightarrow c = \frac{\varepsilon_0 a}{\alpha}[\ln(d + \alpha x)]_0^a$$

$$= \frac{\varepsilon_0 a}{\alpha}\ln\left(1 + \frac{\alpha a}{d}\right) \approx \frac{\varepsilon_0 a^2}{d}\left(1 - \frac{\alpha a}{2d}\right)$$

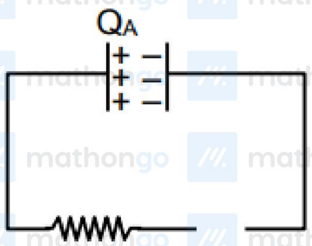


Q12

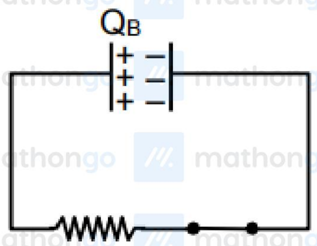
Maximum charge on capacitor $= 5CV$

(a)) is reverse biased and (b) is forward biased

for case (a)



$Q_A = CV$



so $q = q_{max}\,[e^{-t/RC}]$

$Q_B = CVe^{-1}$